

The background features a dark blue gradient with faint, light blue circular patterns and a ruler scale on the left side. The ruler scale has markings from 140 to 260 in increments of 10. Several circular arrows, some solid and some dashed, are scattered across the background, suggesting a circular or cyclical process.

EFFECT OF BEER CONSUMPTION ON REACTION TIME IN ATHLETES: A RULER TEST ANALYSIS

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Research Project

INTRODUCTION

- **Research Question:** Does consuming beer before athletic performance affect reaction times, potentially reducing injury severity?
- **Hypothesis:** Athletes who consume beer before performing have slower reaction times due to muscle relaxation, potentially leading to reduced injury severity.

AIM

- **Objective:** Investigate the effect of beer consumption on athletes' reaction times, using the ruler test, and determine whether this could influence injury risk.
- **Variables:**
 - **Explanatory:** Alcohol consumption (beer vs. no beer), Sex (male vs. female)
 - **Response:** Reaction time (Ruler test: before and after consumption)

PARTICIPANTS

Sample Size: 46 participants (23 males, 23 females), aged 20-30, athletes

Random Assignment:
Divided into beer and control groups

Testing Procedure:
Reaction times were measured using the ruler drop test, both before and after beer consumption

METHODS

- **Study Design:** A 2x2 factorial design with two factors:
 1. Alcohol consumption (yes/no)
 2. Sex (male/female)
- **Response Variable:** Reaction time difference (before beer – after beer)
- **Statistical Analysis:** Two-way ANOVA to assess the effect of alcohol and sex on reaction times, plus residual checks for normality

RESULTS

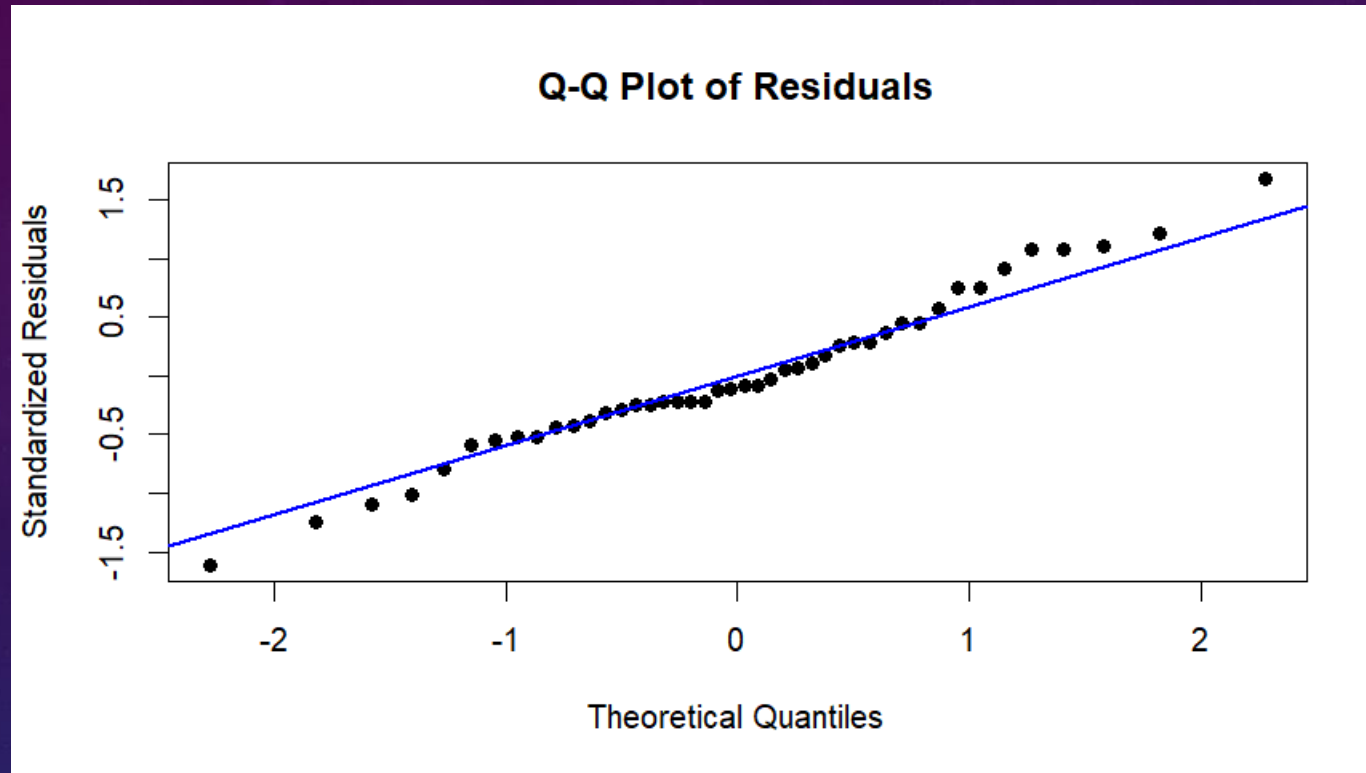
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Alcohol	1	1.566	1.5657	3.089	0.0865
Sex	1	1.2111	1.2111	2.389	0.13
Alcohol:Sex	1	0.46	0.4602	0.908	0.3464
Residuals	40	20.275	0.5069		

- There is no strong evidence that alcohol or sex has a significant effect on reaction times based on the p-values.

Interaction Plot: Reaction Times by Sex and Alcohol Consumption

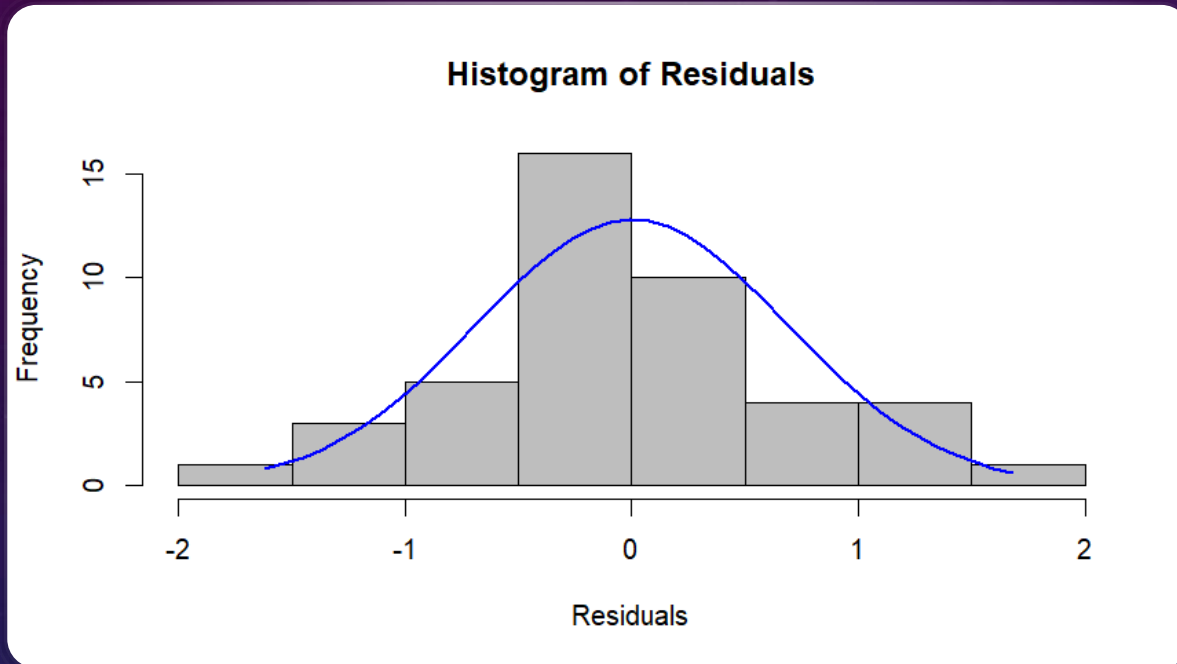


- The lines are nearly parallel
- there is no strong interaction



Residuals are
normally
distributed

No major
deviations
from normality



- The residuals are centered around zero and approximate a normal distribution

CONCLUSION

- **Key Findings:**

1. - No significant effect of alcohol or sex on reaction times (p-values above 0.05).
2. - Small differences observed for males, with alcohol potentially slowing their reaction times more than females.

- **Implications:** These findings suggest that alcohol consumption does not significantly impact athletes' reaction times enough to affect injury risk.

DISCUSSION

- **Limitations:**

- Small sample size
- Only used the ruler test as a measure of reaction time, which may not fully capture the impact of alcohol on injury severity.

- **Future Research:**

- Consider increasing sample size
- Investigate other response variables, such as injury severity or reaction times in real-world sports contexts.

